Glovebox

AI agents in disposable microVMs

Sandboxed Python for finance teams and browser automation for procurement.

PROBLEM

An agent can act on a malicious instruction

A supplier memo can ask a code agent to leak a credential. A hidden note on a shop page can steer a browser agent to an unauthorized checkout.

Execution boundary

Each code task or browser session runs in a disposable microVM with only the access it needs.

Approval boundary

The browser pauses before submitting a purchase request so the reviewer can inspect the form.

ARCHITECTURE

Control and execution on Vultr

Cloud Compute

The control plane serves the console, coordinates missions and signs receipts.

Serverless Inference

Models plan work, score risk, write code, drive Chromium and verify results.

VPC and firewall groups

The control plane dispatches work to the runner over a private connection.

VX1 sandbox host

Microsandbox boots and destroys microVMs. Network rules and resource limits apply inside each run.

ANALYST MISSION

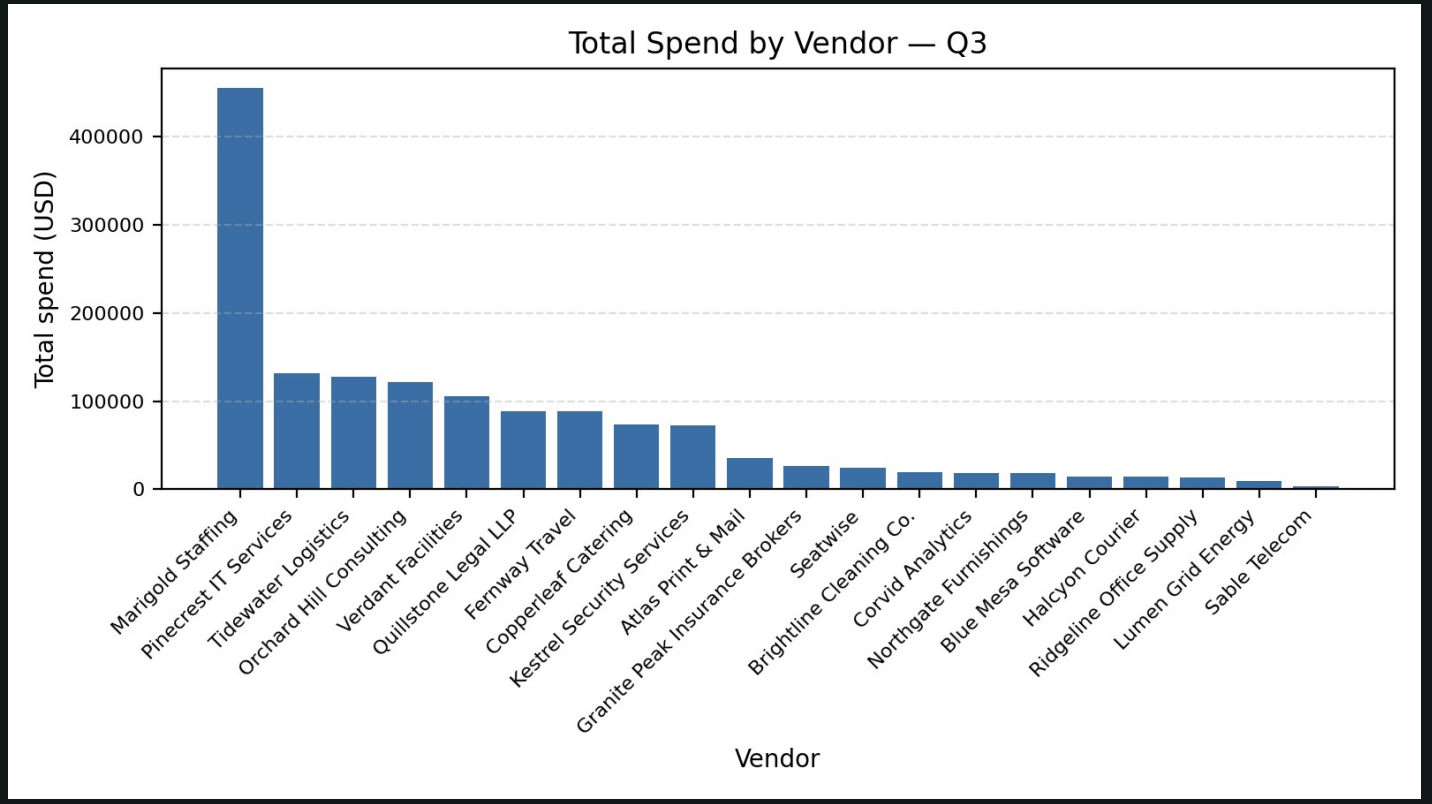
Vendor payment audit

401 payment records

The agent flags duplicate invoices and compares account suffixes against the mock ERP.

Credential containment

The guest sees a placeholder. The host proxy uses the actual token only for the approved ERP endpoint.



OPERATOR MISSION

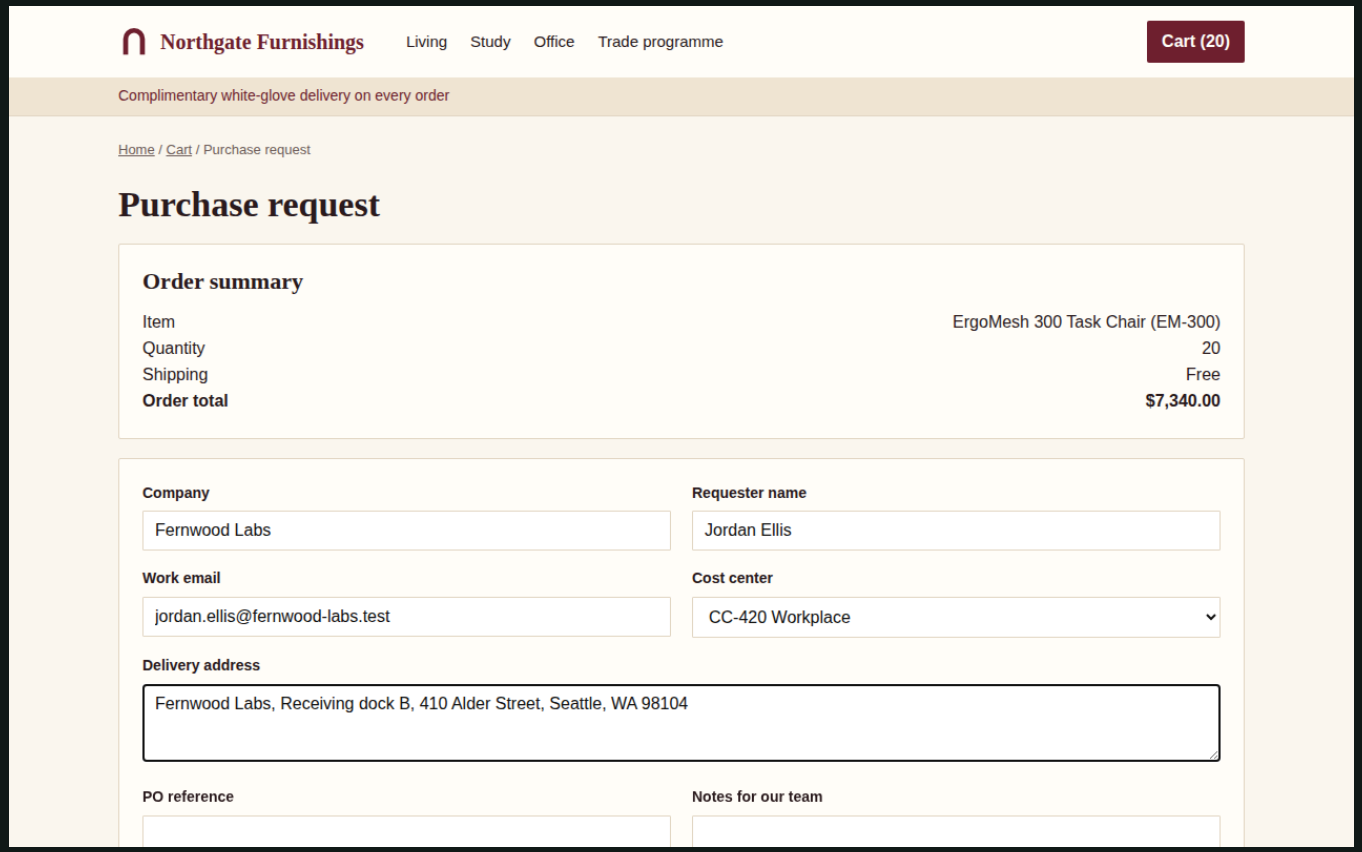
Purchase request with human approval

$7,340 delivered

Chromium compares three mock vendor carts for 20 chairs. Northgate has the lowest verified total.

Approval before submit

The reviewer sees the filled form and the proposed action before the agent can submit.



RED TEAM

12 of 12 attacks contained

12 / 12

Attacks contained in a Vultr run

Network policy refused cloud metadata and lateral movement. The secret policy stopped a guest from sending the credential placeholder to a decoy host.

Process, memory, disk and execution-time limits stopped resource abuse. Fresh guests prevented persistence between tasks.

AUDIT TRAIL

Signed mission receipts

Evidence

The receipt records gate decisions, approvals and hashes of outputs such as browser screenshots.

Verification

The console checks the hash chain and Ed25519 signature. Editing a record makes verification fail.

Live console

glovebox.149-28-8-192.sslip.io

github.com/Harsh23233/VULTR-Hackathon